

The Current MES-Bound Framework and Planck Low-Multipole Results

A self-contained account of the theory, finite-pool statistics,
scalar mechanism audit, and the external-referee correction
of the observer-space irreducible-tensor lane
Critical third revision

HTT / MES research report

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Abstract

This report gives a self-contained statement of the current Maartens–Ellis–Stoeger (MES) bound framework and all presently admitted Planck low-multipole results obtained after the statistical-method upgrade. The theory is a one-way, premise-conditioned ceiling on squared Hubble-normalized shear and vorticity invariants. On the registered geodesic, residual-dipole-free branch, the two ceiling coordinates are deterministic functions of the measured quadrupole and octupole powers. Their nonnegative transformation is invertible and creates neither a sky direction nor independent information. In the exact observation-plus-300 paired CMB-plus-noise pool, $\Sigma_{\max}^2 = 2.40 \times 10^{-10}$ and $W_{\max}^2 = 3.01 \times 10^{-13}$ have local ranks 74/301 each. The frozen ten-coordinate scalar MES-plus-morphology family has rank 98/301 under the legacy leave-one-out absolute-median reducer and 148/301 under the all-two-sided leave-one-out ECDF-midrank reducer. These are conditional finite-pool diagnostics, not physical measurements or unconditional p -values.

The external referee reports exposed a consequential representation defect in the later observer-space lane. The stored harmonic carrier uses the orthonormal real convention $c_{\ell m, c} = \sqrt{2} \Re a_{\ell m}$ and $c_{\ell m, s} = -\sqrt{2} \Im a_{\ell m}$ for $m > 0$, whereas the STF projection and carrier-injection metric interpreted the same entries as unscaled real and imaginary parts. This is not a harmless overall normalization: it treats $m = 0$ and $m > 0$ differently and therefore changes rotations and tensor shape. For example, the same-pass powers require $q_2 = (75/8\pi)C_2 = 628.05 \mu\text{K}_{\text{CMB}}^2$ and $o_2 = (245/8\pi)C_3 = 4698.85 \mu\text{K}_{\text{CMB}}^2$, not the previously reported 1187.75 and 9340.68 $\mu\text{K}_{\text{CMB}}^2$. Consequently the previously quoted observer-space 27/301 result, its 999-row CMB-only comparison, component and foreground replacements, and the carrier-domain power curves are withdrawn from the admitted scientific result set pending correction and complete rerun. Their exact execution and byte/content provenance remain historical evidence; those properties do not restore semantic validity.

The review also correctly establishes that an eight-coordinate catalogue cannot separate the generic nine-dimensional orbit space of a joint $\ell = 2, 3$ pair. It further identifies an overbroad absence rule for a defined zero determinant on noncyclic Krylov strata. These findings are recorded as method defects, not repaired by prose. No current result identifies a physical shear or vorticity, a local/global origin, a foreground source, or a Bianchi family.

Current admission status after external review. The scalar MES and scalar-morphology finite-pool results remain admitted at their existing conditional diagnostic ceiling. Every result that consumes the faulty harmonic-to-STF or injection-basis interpretation is *not currently admitted*: this includes the frame-free and Galactic STF ranks, component-product STF comparisons, foreground-bank projection scores, CMB-only STF ranks, and all WU-008 power statements. This status change is a scientific-semantic correction, not a provenance failure and not a change to a reference value chosen after inspection.

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1 Questions, scope, and logical layers

The analysis addresses three nested questions.

1. **MES premise layer.** Given registered almost-FRW and observer premises, what upper-ceiling coordinates follow from the measured low-order temperature multipole amplitudes?
2. **Observer-space layer.** What radial, shape, mixed, and parity-sensitive information is present in the measured $\ell = 2$ and $\ell = 3$ harmonic blocks themselves?
3. **Finite-pool layer.** How do the observed coordinates rank relative to predeclared, finite Planck simulation ensembles when the complete scoring operator is applied symmetrically to every row?

The arrows between these layers are deliberately one-way:

$$\text{measured sky} \longrightarrow \begin{cases} \text{scalar powers} \longrightarrow \text{MES ceilings,} \\ \text{harmonic blocks} \longrightarrow \text{observer-space STF tensors,} \\ \longrightarrow \text{finite-pool diagnostics.} \end{cases}$$

There is no arrow in this report from a scalar ceiling or an observer-space tensor to a physical spacetime state. Such an inverse would require a separately specified forward response, compatible frame and units, sufficient response rank, and additional observational support.

Table 1 separates the present theoretical and data-analysis results from work that has not been performed.

Table 1: Current result map after the external-referee correction.

Object	Status	Exact scope or ceiling
MES ceiling coordinates	Registered conditional formula	Geodesic, linear almost-FRW, registered frame; residual cosmological $\epsilon_1 = 0$ for the principal branch; full derivative-hierarchy derivation is not reproduced here
Scalar-to-vector/STF lemma	Valid structural lemma	Rotational scalars alone under an $O(3)$ -equivariant construction cannot produce a nonzero vector or STF tensor
Finite-rank validity	Conditional theorem	Joint exchangeability under the declared null plus row-permutation equivariance; conservative ties
Local/global identifiability	Structural result	Full nuisance-projected directional and depth-response rank is required
Scalar Planck analysis	Executed	One SMICA row plus exact paired 300-row CMB-plus-noise pool
Observer-space STF primary	Withdrawn pending rerun	The stored orthonormal real carrier was consumed with the wrong $m > 0$ normalization; the eight-coordinate family is also demonstrably non-separating
CMB-only STF lane	Withdrawn pending rerun	Inherits the same projection defect and, independently, is not exchangeable with the noisy observation
PR3 component and external holdout STF lanes	Withdrawn pending rerun	All consume the faulty projection; product replacement also lacks matched product-specific null ensembles
Foreground projection test	Withdrawn pending redesign	Inherits the faulty projection and has no demonstrated power for subdominant residuals
Carrier-injection power study	Withdrawn pending rerun	The intervention metric and rotations used the unscaled complex-component convention on a $\sqrt{2}$ -scaled real carrier
Physical response inversion	Not performed	No shear, vorticity, local/global, or family inference

2 External-referee adjudication and semantic correction

The two referee reports were not adopted as an all-or-nothing verdict. Each objection was checked against the declared mathematics, the stored numerical objects, and the exact representation contracts. This section records the findings that alter the current result set, the findings that sharpen only the claim ceiling, and the proposed remedies that would require a new experiment.

2.1 The harmonic-basis normalization mismatch

The portable carrier stores a real orthonormal harmonic vector. For each $m > 0$ its entries are

$$c_{\ell m, c} = \sqrt{2} \Re a_{\ell m}, \quad c_{\ell m, s} = -\sqrt{2} \Im a_{\ell m}, \quad c_{\ell 0} = a_{\ell 0}. \quad (1)$$

Hence

$$\|c_\ell\|^2 = a_{\ell 0}^2 + 2 \sum_{m=1}^{\ell} |a_{\ell m}|^2 = (2\ell + 1)C_\ell. \quad (2)$$

The printed STF formulae in Appendix A, however, take the unscaled tuple $(a_{\ell 0}, \Re a_{\ell 1}, \Im a_{\ell 1}, \dots)$ as their input. The required adapter is therefore

$$\Re a_{\ell m} = c_{\ell m, c}/\sqrt{2}, \quad \Im a_{\ell m} = -c_{\ell m, s}/\sqrt{2}. \quad (3)$$

The executed projection omitted Eq. (3). The $m = 0$ entry was unchanged while every $m > 0$ entry was interpreted too large by $\sqrt{2}$ (with the sine-sign convention also lost). This is therefore not a single radial rescaling that would cancel after normalization. It changes the representative under rotation, its eigenstructure, and every mixed Q/O coordinate.

The radial invariants provide an exact internal check. Under the stated STF normalization,

$$q_2 = Q_{ab}Q^{ab} = \frac{15}{8\pi} \left(a_{20}^2 + 2 \sum_{m=1}^2 |a_{2m}|^2 \right) = \frac{75}{8\pi} C_2, \quad (4)$$

$$o_2 = O_{abc}O^{abc} = \frac{35}{8\pi} \left(a_{30}^2 + 2 \sum_{m=1}^3 |a_{3m}|^2 \right) = \frac{245}{8\pi} C_3. \quad (5)$$

The same-pass scalar carrier gives $C_2 = 210.461847 \mu\text{K}_{\text{CMB}}^2$ and $C_3 = 482.020811 \mu\text{K}_{\text{CMB}}^2$. Table 2 shows the decisive mismatch.

Table 2: Representation-normalization audit. Values in the last column are historical outputs of the faulty projection, not alternative estimators.

Invariant	Same-pass C_ℓ	Required by Eqs. (4)–(5)	Previously reported
q_2 ($\mu\text{K}_{\text{CMB}}^2$)	210.462	628.051	1187.747
o_2 ($\mu\text{K}_{\text{CMB}}^2$)	482.021	4698.855	9340.678

The carrier-injection implementation made the reciprocal semantic error: it declared an unscaled complex-component metric $\text{diag}(1, 2, 2, \dots)$ while consuming the already orthonormal vector of Eq. (1). Its nominal unit energies, rotation matrices, and amplitudes therefore do not have the intended carrier meaning. Exact row counts, deterministic replay, and matching hashes verify what that code executed; they cannot establish that the wrong representation implements the intended scientific operator.

2.2 Downstream validity disposition

The scalar feature extraction reconstructs the complex harmonics with the inverse of Eq. (1) before evaluating C_ℓ and the scalar morphology coordinates. Those scalar results do not consume the faulty STF projection and remain within their previous conditional claim ceiling. Every Q/O -derived result does consume it and is withdrawn.

Table 3: Effect of the normalization defect on the reported analysis lanes.

Lane	Current status	Reason
Scalar C_ℓ , MES anchors, and scalar morphology	Retained, conditional	Uses the correctly decoded complex harmonics and does not consume Q/O
Paired-300 frame-free and Galactic STF ranks	Withdrawn	Radial, shape, mixed, and parity coordinates change under the correct adapter
999-row CMB-only STF lane	Withdrawn	Same projection defect; additionally the noisy observation is not exchangeable with a noise-free null pool
Component products and external CMB holdouts	Withdrawn	Same projection defect; the replacement products also lack matched product-specific nulls
Foreground-template projection bank	Withdrawn	Same projection defect and no demonstrated sensitivity to subdominant residual amplitudes
Carrier-domain injection power	Withdrawn	Template energy, rotations, and irrep extraction use the wrong carrier convention

No corrected rank is estimated in this report. A valid replacement requires one convention adapter, rotation-covariance and radial-identity regressions, and a complete rerun of every affected row under the already frozen scientific questions. Reusing the old ranks or altering tolerances would be a false repair.

2.3 The catalogue is demonstrably non-separating

The first referee is correct that “global completeness unproven” was too weak. A real $\ell = 2$ STF tensor has five degrees of freedom and a real $\ell = 3$ STF tensor has seven. At a generic pair (Q, O) with trivial $SO(3)$ stabilizer, quotienting the 12-dimensional product by three rotations leaves a nine-dimensional orbit space:

$$\dim[(\text{STF}_2 \oplus \text{STF}_3)/SO(3)]_{\text{generic}} = 5 + 7 - 3 = 9. \quad (6)$$

Eight smooth coordinates cannot separate a generic nine-dimensional manifold. The old frame-free family is therefore a partial diagnostic catalogue, not a complete orbit registry. The shortfall is scientifically important: Q contributes its amplitude and one shape invariant, while the old catalogue gives O only its amplitude as a pure-octupole invariant; octupole shape otherwise enters through mixed contractions.

Multipole vectors provide a standard competing representation of a fixed multipole and make the missing pure- $\ell = 3$ shape information explicit [6, 7, 8]. The three sorted octupole-vector dot products are already present in the scalar analysis, but simply appending them does not by itself prove that a proposed joint catalogue separates the quotient: discrete sign, permutation, parity, and joint-orientation identifications must be handled, and functional independence must be checked. That is a future redesign, not a post-hoc patch to the withdrawn family.

2.4 Critical disposition of the remaining referee findings

Table 4: Disposition of the principal external-review findings.

Finding	Disposition	Consequence in this revision
No power calibration	Partly sustained, now blocked	The later algebraic injection study did add a coherence-sensitive integer-product combiner, but it neither used physical templates nor survived the carrier-convention audit. No current power claim remains.
Family rank adds “no information” beyond one local rank	Overstated	The row-equivariant family rank is a genuine finite-pool multiplicity calibration, not an independent physical observable. An independence approximation based on an effective dimension does not algebraically force the exact 27/301. The old rank is nevertheless withdrawn for the projection defect.
Galactic lower-side behavior	Sustained as an interpretive warning	In the withdrawn output the upper-tail ranks 252/301 and 289/301 correspond, in the absence of ties, to descriptive lower-side counts 50/301 and 13/301. The lower direction was not predeclared and FFP10 contains no residual foreground; no Galactic inference is admitted.
Foreground bank rules out residuals	Sustained	A squared cosine has quadratic response to a small contaminant, and the simultaneous Q/O minimum further reduces power. The old 144/301 cannot rule out subdominant or mixed foreground residuals even apart from the projection defect.
CMB-only ranks are anti-conservative	Sustained in premise, not as a universal sign theorem	Observation noise is absent from the null and exchangeability fails. Extra noise need not move every normalized shape coordinate in a fixed direction, so the sign of every rank shift is not mathematically forced. The lane is withdrawn rather than interpreted.
Replace every typed absence by zero	Partly sustained	Zero amplitude makes $\hat{Q}$ or $\hat{O}$ genuinely undefined and remains a typed absence. When both normalized tensors exist but the Krylov matrix has rank below three, $\det[v, \hat{Q}v, \hat{Q}^2v]$ is defined and equals zero; a future schema should store zero plus a stratum flag instead of aborting.
Absolute-median ECDF/factorial contrasts	Sustained and con-	Strictly increasing positive reparameterizations preserve ECDF ranks. Squaring contrasts and the flat ECDF ϵ_1 curve are algebraic consequences, not empirical discoveries. The remaining five-count mixing contrast is descriptive and is not assigned significance.

Finding	Disposition	Consequence in this revision
Naive binomial error bars are mandatory	Not adopted as stated	The reported fractions are exact ranks on fixed finite pools with resolution $1/N$. A binomial standard error would add an IID super-population model not used by the rank test. Generalization uncertainty and paired rank differences require a separately specified resampling or simulation design; none is fabricated here. Display precision is nevertheless reduced to what interpretation supports.
MES coefficients and comparison literature are underdocumented	Sustained	The carriers are retained as registered premise-conditioned formulae, not promoted as a newly derived theorem. Their derivative-hierarchy premises and the difference from model-dependent Bianchi constraints are stated below.
Program-level selection is hidden	Sustained	Table 8 lists every analysis family, reducer, pool, product, and intervention reported here, including withdrawn lanes and explicitly unexecuted alternatives. Content hashes establish identity, not external preregistration chronology.

3 Cosmological and MES-bound framework

3.1 Kinematics and normalized target invariants

Let u^a be the registered fundamental-observer four-velocity, $u^a u_a = -1$, and

$$h_{ab} = g_{ab} + u_a u_b$$

the spatial projector. With metric signature $(-, +, +, +)$ and $\epsilon_{123} = +1$, the covariant velocity gradient is decomposed as

$$\nabla_b u_a = -u_b A_a + \frac{1}{3} \Theta h_{ab} + \sigma_{ab} + \omega_{ab},$$

where $A_a = u^b \nabla_b u_a$ is the acceleration, $\Theta = \nabla_a u^a$ is the expansion, $\sigma_{ab} = \sigma_{(ab)}$ is the spatial shear, and $\omega_{ab} = \omega_{[ab]}$ is the spatial vorticity. Define $H = \Theta/3$. The current MES coordinates target the squared dimensionless invariants

$$\Sigma^2 \equiv \frac{\sigma_{ab} \sigma^{ab}}{6H^2}, \quad W^2 \equiv \frac{\omega_{ab} \omega^{ab}}{6H^2}. \quad (7)$$

These are nonnegative scalar magnitudes. They are not vectors, principal axes, eigenframes, or signed tensors.

The MES programme uses the near-isotropy of the cosmic microwave background to place premise-dependent constraints on departures from an almost-FRW state [1, 2]. The present version uses the geodesic branch $A_a = 0$, the linear almost-EGS regime, and a registered fundamental-observer frame. Coefficients involving time or spatial derivatives of radiation multipoles inherit the corresponding derivative- hierarchy premises; they are not universal identities of arbitrary spacetimes.

3.2 Temperature multipole amplitudes

For a temperature anisotropy field with angular power C_ℓ , define

$$\epsilon_\ell = \frac{1}{T_0} \sqrt{\frac{(2\ell+1)C_\ell}{4\pi}}, \quad (8)$$

where T_0 is expressed in the same temperature unit as $\sqrt{C_\ell}$. The reported numerical analysis fixes $T_0 = 2.72548 \text{ K} = 2.72548 \times 10^6 \mu\text{K}_{\text{CMB}}$. Thus every ϵ_ℓ is dimensionless and nonnegative. The observed dipole is dominated by observer motion. In the principal branch used below, that kinematic contribution is assigned to the observer-motion channel and the residual cosmological dipole is set to

$$\epsilon_1 = 0.$$

This is an explicit premise, not a conclusion extracted from the quadrupole or octupole.

3.3 Latest MES carriers and one-way ceilings

The registered geodesic carriers, before imposing $\epsilon_1 = 0$, are

$$B_\sigma = \frac{5}{3}\epsilon_1 + 3\epsilon_2 + \frac{3}{7}\epsilon_3, \quad (9)$$

$$B_\omega = \frac{10}{3}\epsilon_1 + \frac{2}{15}\epsilon_2. \quad (10)$$

On the principal branch they reduce to

$$B_\sigma = 3\epsilon_2 + \frac{3}{7}\epsilon_3, \quad B_\omega = \frac{2}{15}\epsilon_2.$$

The squared normalized ceiling coordinates are

$$\Sigma^2 \leq \Sigma_{\text{max}}^2 = \frac{3}{2}B_\sigma^2 = \frac{27}{98}(7\epsilon_2 + \epsilon_3)^2, \quad (11)$$

$$W^2 \leq W_{\text{max}}^2 = \frac{3}{2}B_\omega^2 = \frac{C_2}{30\pi T_0^2}. \quad (12)$$

The square symbols are essential: the reported variables are ceilings on the squared invariants in Eq. (7), not unsquared shear or vorticity amplitudes.

Equations (9)–(10) are the recovered, registered carriers used by the scalar analysis. This report does not reproduce the full covariant radiation-multipole derivation that fixes every derivative term and then justifies each omission. They must therefore be read as conditional formulae under the stated geodesic, linear almost-EGS, derivative-hierarchy, frame, and dipole premises, rather than as universal identities of a measured sky. The algebra from those carriers to Eqs. (11)–(12) is exact; the physical scope of the carriers remains premise-dependent.

These inequalities are one-way. A sky realization produces ceiling coordinates after the premises have been declared. The converse—that a ceiling determines a unique physical source, geometry, direction, or spacetime—does not follow.

The scientific role also differs from model-dependent Bianchi inference. Full forward-model analyses of Planck temperature and polarization constrain particular anisotropic-expansion modes much more tightly—for example a vector-mode limit $(\sigma_V/H)_0 < 4.7 \times 10^{-11}$ and a weaker regular tensor-mode limit of order 10^{-6} were reported by Saadeh et al. [9]. Those mode amplitudes, priors, transfer functions, and normalizations are not identical to Eq. (7), so the numbers must not be divided as if they were the same parameter. The purpose of the MES ceiling here is its transparent one-way premise structure, not competitive numerical strength or a replacement for a physical likelihood.

3.4 Invertibility and absence of information gain

On the registered nonnegative domain the principal-branch transformation can be inverted:

$$\epsilon_2 = \sqrt{\frac{75}{2} W_{\max}^2}, \quad (13)$$

$$\epsilon_3 = \sqrt{\frac{98}{27} \Sigma_{\max}^2} - 7\epsilon_2, \quad (14)$$

$$C_\ell = \frac{4\pi T_0^2}{2\ell + 1} \epsilon_\ell^2. \quad (15)$$

Consequently, for any accompanying morphology vector M ,

$$\sigma(\Sigma_{\max}^2, W_{\max}^2, M) = \sigma(C_2, C_3, M) \quad (16)$$

on this domain. The MES coordinates can expose a useful physical normalization and can change the geometry of a chosen rank reducer, but they do not add statistically independent information to the same-row C_2, C_3 inputs.

3.5 Why a scalar ceiling cannot create direction or STF shape

Suppose s denotes any collection of rotation-invariant scalars and $F(s) \in \mathbb{R}^3$ is $O(3)$ -equivariant. Since $Rs = s$ for every rotation R ,

$$F(s) = F(Rs) = RF(s) \quad \text{for all } R \in O(3).$$

The only vector fixed by every rotation is zero. The same argument in the irreducible rank-two STF representation shows that an equivariant map from scalars alone to an STF tensor must also vanish. Therefore

scalar MES ceiling $\not\Rightarrow$ direction, nonzero vector, or nonzero STF tensor.

Direction-indexed data do permit legal moment maps. For a full-sky scalar field $q(\hat{n}, z)$,

$$V_a(z) = \frac{3}{4\pi} \int q(\hat{n}, z) n_a d\Omega, \quad (17)$$

$$T_{ab}(z) = \frac{15}{8\pi} \int q(\hat{n}, z) n_{\langle a} n_{b \rangle} d\Omega. \quad (18)$$

If $q = v_a n^a + S_{ab} n^a n^b$ plus harmonics orthogonal to $\ell = 1, 2$, with S_{ab} STF, these normalizations recover $V_a = v_a$ and $T_{ab} = S_{ab}$. For a masked or discretized sky, the corresponding estimator and quadrature must be bound explicitly. A single global scalar does not meet this premise.

3.6 Response-bound states and local/global identifiability

The current framework distinguishes three semantic objects:

Observable irrep state

Measured or simulated harmonic/STF content with a declared frame, basis, units, operator, and typed degenerate strata.

MES normalizer

A positive, same-channel, one-way ceiling that may rescale an already existing compatible amplitude. It creates no direction, shape, response, or information.

Response-bound state

An observable state tied to an independently specified forward response with compatible channel, frame, basis, units, epoch, and normalization. Its response rank determines point or set identification.

A rank-deficient response yields an identified set, not a point estimate. A simple local/global directional model illustrates the requirement:

$$V_i = f_L(z_i) \beta_L + f_G(z_i) \beta_G + \varepsilon_i. \quad (19)$$

Let S be the directional metric and K the 2×2 covariance-weighted depth Gram matrix for f_L, f_G after nuisance projection. The six-parameter Fisher matrix is $K \otimes S$, so

$$\det F = \det(K)^3 \det(S)^2. \quad (20)$$

One shell makes K rank one; proportional depth profiles do the same. Planck temperature morphology alone therefore cannot distinguish local boost from global tilt. No such physical inversion is reported below.

4 Data objects, common operator, and finite-pool statistics

4.1 Observation and two distinct null ensembles

The observation is the Planck PR3 SMICA temperature map. SMICA is a component-separated Planck product [3]; the large-angle statistical context and its sensitivity to masks and foreground treatment are described by the Planck isotropy analysis [4].

Two null ensembles have different scientific roles.

1. **Paired primary.** Exactly 300 ordered FFP10 SMICA CMB-plus-noise realizations, numbered 00000–00299, combined with the single observation to give $N = 301$ rows. This is the primary finite pool.
2. **Withdrawn CMB-only STF lane.** Exactly 999 official FFP10 SMICA CMB-only realizations, numbered 00000–00999 with only 00970 absent and 00818 included, combined with the same observation to give $N = 1000$ rows. No noise realization enters this lane. It does not replace the paired primary; after review it is not an admitted calibrated test because the noisy observation and noise-free rows are not jointly exchangeable, independently of the STF representation defect.

The common temperature operator performs one joint masked real-harmonic fit over $\ell = 0, \dots, 5$ and retains the 32 real coefficients at $\ell = 2, \dots, 5$. The registered frame is Galactic, the map ordering is HEALPix RING, the processing grid is $N_{\text{side}} = 16$, and the unit is μK_{CMB} . The mask is the Planck PR3 common temperature mask, with frozen content identity `sha256:34390a5c2d9c1462af87d51a9a3e88971781b23054d923563723531b7f30ea94`. The portable artifact does not expose an independently checkable f_{sky} scalar, so no value is invented here. Source beam and pixel window are mapped to the declared common target before retained-feature evaluation. The observed carrier was obtained in the same pass as its scalar features; it was neither reconstructed from those scalars nor estimated a second time. Its scalar closure matches the frozen same-pass values, and the operator condition number is 6.2075, far below the predeclared ceiling 10^8 ; a condition number at or above that ceiling would have aborted the fit.

No separate kinematic-quadrupole or aberration correction was introduced in this frozen operator. Observation and simulations therefore passed through the same estimator code, but that symmetry does not imply that every physical observer-motion contribution is represented identically by the simulation products. This is an operator premise, not a validated kinematic correction.

Exact row identity and deterministic replay establish what was compared; they do not establish that either simulation pool is a complete physical model of the observation.

4.2 Reducer A: leave-one-out absolute median

Let X_{ij} be coordinate j for row i . For a two-sided coordinate,

$$d_{ij} = |X_{ij} - \text{median}(X_{-i,j})|. \quad (21)$$

For an upper tail use $d_{ij} = X_{ij}$ and for a lower tail use $d_{ij} = -X_{ij}$. The observation-inclusive local rank is

$$p_{ij} = \frac{1}{N} \sum_{k=0}^{N-1} \mathbf{1}(d_{kj} \geq d_{ij}). \quad (22)$$

Because every row has its own leave-one-out centre, the complete map from all rows to all scores, not only the observation-row formula, is the statistical object. This reducer is invariant under positive affine rescaling but not under arbitrary monotone reparameterization.

4.3 Reducer B: leave-one-out ECDF midrank

For every row and coordinate define

$$u_{ij} = \frac{\#\{k \neq i : X_{kj} < X_{ij}\} + \frac{1}{2}\#\{k \neq i : X_{kj} = X_{ij}\}}{N-1}. \quad (23)$$

The oriented score is $2|u_{ij} - 1/2|$ for a two-sided coordinate, u_{ij} for an upper tail, and $1 - u_{ij}$ for a lower tail. Exact integer midrank numerators preserve mathematical ties. This reducer is invariant under strictly increasing coordinate transformations that preserve tie structure.

4.4 Family scan, ties, and the finite-rank premise

For a predeclared family $\mathcal{F}$, every row receives

$$r_i = \min_{j \in \mathcal{F}} p_{ij}, \quad S_i = -\log r_i. \quad (24)$$

The reported family rank is

$$p_{\mathcal{F}} = \frac{1}{N} \sum_{k=0}^{N-1} \mathbf{1}(r_k \leq r_0) = \frac{1}{N} \sum_{k=0}^{N-1} \mathbf{1}(S_k \geq S_0). \quad (25)$$

Ties and the observation itself are included, so a zero rank is impossible; the resolution floors are $1/301$ and $1/1000$ [5].

If $(Z_0, \dots, Z_{N-1})$ are jointly exchangeable under the stated null and the complete row-scoring map is permutation-equivariant, then the observation-inclusive family rank is super-uniform, with ties treated conservatively. Row-swap and row-permutation checks establish the second condition. The first condition is a scientific null-model premise. If FFP10 does not faithfully represent the observation under the intended null, the fractions remain exact ranks relative to the frozen pool but do not acquire unconditional Type-I-error calibration.

The proof is short. Exchangeability of the input rows and equivariance of the complete map imply exchangeability of $(S_0, \dots, S_{N-1})$. Conditional on their unordered multiset, the observation label is uniformly distributed over the N positions. Counting all scores at least as large as S_0 , including ties, can only increase the ordinary randomized rank; hence the reported rank is super-uniform.

Feature families, coordinate transformations, tails, reducers, missing-row rules, and the ϵ_1 grid were fixed before observation-row scoring. No covariance matrix, Gaussian likelihood, posterior, Bayes factor, or observation-selected feature subset enters Eq. (25).

5 Scalar MES and morphology results

5.1 The eight non-anchor morphology coordinates

The scalar analysis combines two low-order amplitude or MES coordinates with eight morphology coordinates. Define the even-to-odd power ratio

$$\mathcal{P} = \frac{C_2 + C_4}{C_3 + C_5}. \quad (26)$$

For each ℓ , form the real symmetric angular-momentum power tensor

$$\tilde{A}_{ij}^{(\ell)} = \frac{1}{2\ell(\ell+1)} \sum_{m,m'} a_{\ell m}^* \{J_i, J_j\}_{mm'} a_{\ell m'}, \quad A^{(\ell)} = \frac{\tilde{A}^{(\ell)}}{\text{tr } \tilde{A}^{(\ell)}}. \quad (27)$$

If $\lambda_{\ell,1} \leq \lambda_{\ell,2} \leq \lambda_{\ell,3}$ are its trace-one eigenvalues, the registered gaps are

$$G_\ell = \lambda_{\ell,3} - \lambda_{\ell,2}, \quad \ell = 2, 3.$$

Multipole vectors are the headless axes in the harmonic-polynomial factorization

$$T_\ell(\hat{n}) = A_\ell \left[\prod_{r=1}^{\ell} (\mathbf{v}_{\ell,r} \cdot \hat{n}) - \text{traces} \right], \quad (28)$$

defined up to permutation and individual sign changes [6, 7]. Sign-invariant coordinates are

$$d_2 = |\mathbf{v}_{2,1} \cdot \mathbf{v}_{2,2}|, \\ (d_{3,0}, d_{3,1}, d_{3,2}) = \text{sort}(|\mathbf{v}_{3,1} \cdot \mathbf{v}_{3,2}|, |\mathbf{v}_{3,1} \cdot \mathbf{v}_{3,3}|, |\mathbf{v}_{3,2} \cdot \mathbf{v}_{3,3}|).$$

Let $\hat{\mathbf{n}}_2$ be the normalized quadrupole plane normal and $\hat{\mathbf{n}}_{3,ij}$ the three normalized octupole pair normals. The plane-alignment coordinate is

$$A_{23} = \max_{i < j} |\hat{\mathbf{n}}_2 \cdot \hat{\mathbf{n}}_{3,ij}|. \quad (29)$$

The eight-coordinate morphology block is therefore $M = (\mathcal{P}, G_2, G_3, d_2, d_{3,0}, d_{3,1}, d_{3,2}, A_{23})$.

5.2 Six paired family decompositions

Six families were compared on the identical 301 rows with the original all-two-sided, leave-one-out absolute-median operator:

Family	First coordinates	Remaining coordinates
Generic 12	C_2, C_3, C_4, C_5	M
Raw reduced 10	C_2, C_3	M
Linear- ϵ 10	ϵ_2, ϵ_3	M
MES-squared 10	$\Sigma_{\max}^2, W_{\max}^2$	M
Anchors only 2	$\Sigma_{\max}^2, W_{\max}^2$	none
Morphology only 8	none	M

Table 5: Exact scalar family ranks in the paired 300-row primary pool. The families are a paired operator decomposition, not six independent tests.

Family	Global rank	Observation-row minimum
Generic 12	133/301	$d_{3,0} : 16/301$
Raw reduced 10	110/301	$d_{3,0} : 16/301$
Linear- ϵ 10	109/301	$d_{3,0} : 16/301$
MES-squared 10	98/301	$d_{3,0} : 16/301$
Anchors only 2	78/301	both anchors: 74/301
Morphology only 8	88/301	$d_{3,0} : 16/301$

Table 6: Complete observed MES-squared family coordinates and paired-null summaries. Quantiles are the linear 5th, 50th, and 95th percentiles of the 300 frozen null rows; they are descriptive and do not alter the rank operator.

Coordinate	Observed	Local rank	Null median	Null 5–95%
$\Sigma_{\max}^2$	2.40×10^{-10}	74/301	7.86×10^{-10}	$(2.50 \times 10^{-10}, 2.06 \times 10^{-9})$
$W_{\max}^2$	3.01×10^{-13}	74/301	1.23×10^{-12}	$(3.23 \times 10^{-13}, 3.48 \times 10^{-12})$
$\mathcal{P}$	0.553	95/301	1.698	(0.585, 5.959)
G_2	0.450	182/301	0.389	(0.144, 0.491)
G_3	0.560	35/301	0.329	(0.086, 0.553)
d_2	0.133	177/301	0.269	(0.027, 0.727)
$d_{3,0}$	0.399	16/301	0.115	(0.012, 0.396)
$d_{3,1}$	0.434	155/301	0.313	(0.094, 0.583)
$d_{3,2}$	0.437	174/301	0.553	(0.222, 0.879)
A_{23}	0.940	135/301	0.810	(0.391, 0.980)

The quadrupole has only one independent shape degree of freedom after its amplitude and rotation are removed. Consequently G_2 , d_2 , and the STF shape invariant J_Q used in the withdrawn tensor lane are alternative, partly many-to-one parameterizations of the same generic quadrupole shape, not independent evidence channels. Their simultaneous appearance in different sensitivity families is a coordinate-choice audit and must not be counted as multiplicity-free information.

The observed squared ceilings are

$$\Sigma_{\max}^2 = 2.4000870222 \times 10^{-10}, \quad W_{\max}^2 = 3.0061887931 \times 10^{-13}. \quad (30)$$

They follow from

$$C_2 = 210.4618472 \mu\text{K}_{\text{CMB}}^2, \quad C_3 = 482.0208114 \mu\text{K}_{\text{CMB}}^2,$$

which give $\epsilon_2 = 3.3575598243 \times 10^{-6}$ and $\epsilon_3 = 6.0122078079 \times 10^{-6}$ through Eq. (8). Both have local rank 74/301. Their Pearson correlation over the complete pool is 0.994874, their Spearman correlation is 0.993738, and the two-coordinate correlation block has condition number 389.2. These facts quantify strong same-row dependence; they are not evidence for two independent measurements.

Thus the statement supported by the data is narrow: the observation-row minimum is morphology-driven rather than anchor-driven. The complete family rank nevertheless depends on the coordinate family because adding or replacing coordinates changes the row-score distribution for every null row.

5.3 Factorial coordinate-mechanism audit

To separate nonlinear squaring from carrier mixing, four ten-coordinate cells were fixed on the same 301 rows:

$$\begin{aligned} \text{linear } \epsilon &: (\epsilon_2, \epsilon_3, M), \\ \text{square only} &: (\epsilon_2^2, \epsilon_3^2, M), \\ \text{carrier only} &: (B_\sigma, B_\omega, M), \\ \text{MES squared} &: (\tfrac{3}{2}B_\sigma^2, \tfrac{3}{2}B_\omega^2, M). \end{aligned}$$

Table 7: Factorial family ranks under the two registered reducers and the all-two-sided tail registry.

Cell	Absolute-median rank	ECDF-midrank rank
Linear ϵ	109/301	153/301
Square only	110/301	153/301
Carrier only	98/301	148/301
MES squared	98/301	148/301

For the ECDF reducer, the equality of the linear and squared cells is not an empirical discovery: $x \mapsto x^2$ and $B \mapsto \frac{3}{2}B^2$ are strictly increasing on the registered nonnegative domain and therefore preserve ECDF ranks and ties. Only the carrier-mixing contrast remains non-tautological, changing 153/301 to 148/301. No bootstrap or super-population uncertainty was predeclared for that paired five-count difference, so it is retained only as a descriptive coordinate-mechanism sensitivity. Under the absolute-median reducer the nonlinear change of metric can itself alter ranks; its 98/301 value is partly a property of that coordinate mechanism and reducer.

5.4 Tail-registry sensitivity

The factorial comparison above retains the all-two-sided tail convention so that coordinate-mechanism effects are isolated. A separate, predeclared physics-oriented registry uses upper tails for the two MES ceilings, the two power-tensor gaps, and the plane-alignment maximum, while retaining two-sided tails for parity and all four multipole-vector dot products. Applied to the MES-squared cell, the results are

Tail registry	Absolute median	ECDF midrank
All two-sided	98/301	148/301
Physics-oriented	98/301	93/301

The unchanged absolute-median rank and changed ECDF rank are another paired operator-sensitivity result. Neither row is an independently selected test, and the 93/301 value does not supersede the all-two-sided factorial result.

5.5 Residual-dipole premise sensitivity

Equations (9)–(10) were evaluated rowwise on a predeclared 101-point grid $0 \leq \epsilon_1 \leq 10^{-5}$. The observed ceilings change substantially over that interval: at the upper endpoint, $\Sigma_{\max}^2$ is 5.37 times and $W_{\max}^2$ is 5694 times their $\epsilon_1 = 0$ values. Nevertheless the complete MES-squared family ranks remain 98/301 and 148/301 throughout the grid for the two reducers.

For ECDF this constancy is forced by the same monotone-transformation theorem: adding the same nonnegative dipole term to every row and then squaring remains strictly increasing over this grid. It is not evidence that vorticity is insensitive to a residual cosmological dipole. The absolute ceiling changes by orders of magnitude and the grid is only a bounded premise stress test; it is not a data-derived interval for intrinsic ϵ_1 .

6 Withdrawn observer-space irreducible-tensor lane

Every numerical rank in this section and in the following component, CMB-only, and power sections is retained only to document the superseded execution. None belongs to the current admitted scientific result set because of the normalization mismatch in Section 2.

6.1 Harmonic convention and reversible STF bridge

For a real temperature map the retained convention is

$$T_\ell(\hat{n}) = a_{\ell 0} Y_{\ell 0}(\hat{n}) + 2 \sum_{m=1}^{\ell} \Re[a_{\ell m} Y_{\ell m}(\hat{n})], \quad (31)$$

with orthonormal Condon–Shortley harmonics. The real storage block is the orthonormal vector in Eq. (1), not the unscaled tuple $(a_{\ell 0}, \Re a_{\ell 1}, \Im a_{\ell 1}, \dots)$. The executed bridge failed to apply Eq. (3) before the STF formulae.

The $\ell = 2$ block is mapped linearly to a real symmetric trace-free tensor Q_{ab} , and the $\ell = 3$ block to a real fully symmetric trace-free tensor O_{abc} [10]. Five and seven independent Cartesian components respectively match the dimensions $2\ell + 1$. The intended normalization is given in Appendix A. Every executed row passed

$$a_{\ell m} \longrightarrow (Q, O) \longrightarrow a_{\ell m}$$

with maximum absolute residual $1.4210854715 \times 10^{-14}$ against the predeclared $\text{atol} = \text{rtol} = 5 \times 10^{-13}$. That round trip validated a map and its own inverse under the same wrong input convention. It did not validate the carrier-to-formula adapter or rotation covariance and is therefore not evidence for semantic correctness.

6.2 Radial, shape, mixed, and parity coordinates

The radial and quadrupole-shape invariants are

$$q_2 = Q_{ab} Q^{ab}, \quad q_3 = Q_a^{\ b} Q_b^{\ c} Q_c^{\ a}, \quad (32)$$

$$o_2 = O_{abc} O^{abc}, \quad J_Q = \sqrt{6} \frac{q_3}{q_2^{3/2}}. \quad (33)$$

Here q_2, o_2 have units $\mu\text{K}_{\text{CMB}}^2$, q_3 has units $\mu\text{K}_{\text{CMB}}^3$, and J_Q is dimensionless. For a real symmetric trace-free 3×3 tensor,

$$Q^3 - \frac{q_2}{2} Q - \frac{q_3}{3} I = 0, \quad \frac{q_2^3}{2} - 3q_3^2 \geq 0,$$

so $J_Q \in [-1, 1]$ when $q_2 > 0$.

When $q_2, o_2 > 0$, define

$$\hat{Q} = Q/\sqrt{q_2}, \quad \hat{O} = O/\sqrt{o_2}, \quad v_a = \hat{O}_{abc} \hat{Q}^{bc},$$

and

$$S_{ab} = \text{STF}(\hat{O}_{acd} \hat{O}_b^{\ cd}).$$

The registered dimensionless mixed coordinates are

$$\hat{R}_{v0} = v_a v^a, \quad \hat{R}_{v1} = v_a \hat{Q}^a{}_b v^b, \quad (34)$$

$$\hat{R}_{v2} = v_a (\hat{Q}^2)^a{}_b v^b, \quad \hat{R}_{QS} = \hat{Q}_{ab} S^{ab}, \quad (35)$$

$$\hat{K}_v = \det[v, \hat{Q}v, \hat{Q}^2 v]. \quad (36)$$

The first four are reflection-even; $\hat{K}_v$ is a pseudoscalar that changes sign under reflection. It is therefore an $SO(3)$ invariant with explicit parity covariance, not an ordinary $O(3)$ scalar.

The frame-free family is

$$\mathcal{F}_{\text{ff}} = \{q_2, o_2, J_Q, \hat{R}_{v0}, \hat{R}_{v1}, \hat{R}_{v2}, \hat{R}_{QS}, \hat{K}_v\}. \quad (37)$$

For a separate absolute-frame sensitivity check, let $z_G = (0, 0, 1)$ be the Galactic pole and define

$$P_Q^G = |\hat{Q}z_G|^2, \quad P_O^G = |\hat{O}_{abc} z_G^b z_G^c|^2.$$

The ten-coordinate companion is $\mathcal{F}_G = \mathcal{F}_{\text{ff}} \cup \{P_Q^G, P_O^G\}$. It is descriptive and is not a physical source-direction estimator.

6.3 Degenerate strata and typed absence

At $q_2 = 0$ the quadrupole shape is undefined; at $o_2 = 0$ the octupole shape and normalized mixed coordinates are undefined. Those are genuine typed absences: no fallback axis or silent zero can repair a missing normalized tensor. A repeated quadrupole spectrum likewise makes a distinguished eigenaxis unavailable when a coordinate explicitly requires that axis.

The determinant

$$\hat{K}_v = \det[v, \hat{Q}v, \hat{Q}^2 v]$$

has a different status. Once $\hat{Q}$ and $\hat{O}$ exist, it is a polynomial scalar and remains defined when its Krylov matrix has rank below three; on that stratum its value is exactly zero. Treating every such case as absence is therefore overbroad. A corrected schema must store the value zero together with an explicit noncyclic-stratum flag. The old executions happened to contain no row that triggered this branch, but that empirical fact does not make the type rule sound.

Completeness is not merely unproven: Eq. (6) proves that the eight-coordinate family cannot generically separate the nine-dimensional joint orbit space.

7 Withdrawn executions: historical scope without scientific admission

The faulty bridge was consumed by four completed execution groups: the paired 300-row STF analysis, the 999-row CMB-only comparison, component/holdout and foreground replacements, and the observation-free carrier-injection study. Their row graphs, logs, and content identities remain useful for determining exactly what ran, but their numerical ranks and power curves are not scientific results under the intended representation. They are therefore omitted from the active tables and figures of this revision rather than being visually watermarked as if they were alternative estimates.

The historical outputs included a paired family rank 27/301, CMB-only ranks 61/1000 and 55/1000, component-product ranks, a foreground-bank rank 144/301, and 863,040 positive carrier injections. These numbers are stated only to identify the withdrawn objects. None may be reused, combined, or interpreted until the carrier adapter, rotation covariance, radial identities, and full affected

row sets have been rerun. In particular, exact replay of the old bytes demonstrates execution provenance, not semantic equivalence to the intended operator.

The CMB-only lane also has an independent scientific defect: a noisy observed map is not jointly exchangeable with a noise-free null ensemble. The component and cross-instrument replacements lack matched product-specific nulls, while the foreground score has no established sensitivity to small or mixed residuals. Correcting the basis convention alone would therefore not promote those lanes automatically.

8 Analysis-program inventory and chronology

Table 8 makes the selection surface explicit. A row marked “retained” is in the present result set; “withdrawn” means it was executed but fails the current semantic or null premise; “not executed” is neither a null result nor evidence of absence.

Table 8: Program-level inventory after external-referee adjudication.

Analysis lane	Fixed choices and extent	Status	Chronology or bound- ary
Scalar six-family decom- position	One observation plus 300 paired nulls; six declared co- ordinate families; absolute- median reducer	Retained, con- ditional	Precedes the STF ex- tension; rank table frozen
Scalar reducer and tail audit	4×2 coordinate- mechanism/reducer cells; two tail registries	Retained, de- scriptive	ECDF squaring iden- tities are theorems; the five-count mixing shift is not assigned significance
Residual-dipole stress	101 fixed ϵ_1 values	Retained as premise stress	Flat ECDF ranks fol- low monotonicity, not empirical robustness
Paired STF and Galactic families	Eight and ten coordinates; two reducers; 301 rows	Withdrawn	Wrong carrier-to-STF adapter
SMICA CMB-only STF	999 nulls plus observation; two reducers	Withdrawn	Same adapter defect and failed exchange- ability premise
Component and CMB holdouts	Four Planck products and three external CMB products	Withdrawn	Same adapter defect; unmatched nulls
Foreground bank	Thirteen fixed templates and one max–min score	Withdrawn	Same adapter defect; weak small-residual response
Carrier intervention	Five algebraic templates, seven amplitudes, 32 orienta- tions, two arms, two reducers and two combiners	Withdrawn	Injection metric used the wrong carrier con- vention; no physical template
Other multipole pairs, ecliptic/dipole compan- ions, physical Bianchi templates, and matched product nulls	No admitted execution	Not executed	Future questions re- quiring a new frozen experiment, not retroactive repairs

The Git chronology establishes only execution order. The scalar/statistical framework was frozen before the observer-space extensions; WU-006 recorded its operator/result sequence on 29 August 2026, and the WU-008 intervention registry preceded its result commits later that day. These repository times are not evidence of external preregistration and are not used to rescue a semantically invalid operator.

9 Synthesis and scientific interpretation

9.1 What changed with the latest framework

The current methodology replaces four potential shortcuts with explicit separations.

1. **Ceiling versus state.** $\Sigma_{\max}^2$ and $W_{\max}^2$ are squared, premise-conditioned upper coordinates. They are neither the physical tensors nor estimates of their directions.
2. **Scalar versus shape.** Directional morphology must come from the measured harmonic blocks through an explicitly verified representation adapter, never from scalar reconstruction. The withdrawn STF lane is a concrete example of why an algebraic round trip alone is insufficient.
3. **Exact rank versus null fidelity.** A finite rank can be arithmetically exact relative to a frozen pool while its unconditional scientific calibration remains limited by exchangeability and simulation fidelity.
4. **Execution identity versus semantic validity.** Exact row order, hashes, replay, and a self-consistent inverse prove what was executed. They do not prove that stored coordinates were interpreted in the scientific convention required by the intended operator.

The scalar audit shows that the MES ceilings themselves are not locally unusual in the paired pool: both are 74/301. The smallest scalar local rank is the octupole morphology coordinate $d_{3,0}$ at 16/301. After scanning the full ten-coordinate scalar family, the observation-inclusive rank is 98/301 for the absolute-median reducer. The observation-row minimum is therefore morphology-driven, while the family rank remains dependent on the complete coordinate family and null-row score distribution.

No admitted observer-space rank remains. The old mixed-coordinate minima, component comparisons, foreground score, CMB-only ranks, and injection-power curves all depend on the faulty basis interpretation. The correct scientific transition is withdrawal followed by a fully corrected rerun, not a verbal qualification of the old numbers. The eight-coordinate catalogue would still remain a partial diagnostic after that rerun unless it were replaced by a demonstrably separating joint-orbit construction.

9.2 What the results do not identify

Correctly constructed quantities Q_{ab} and O_{abc} would be observer-space representations of CMB harmonics. They would not be the spacetime shear tensor, vorticity tensor, anisotropic spatial curvature, or a Bianchi template. No currently admitted Q/O data result exists, and no admitted forward response connects a future corrected orbit coordinate to those physical states. No depth-indexed directional channel supplies the rank required by Eq. (20). Therefore no result here identifies a local boost, global tilt, physical source, or geometry family.

The present analysis also does not produce the separate diagnostic vocabulary (x, Q, Π, F, G) , a likelihood, posterior, evidence ratio, posterior predictive check, or cross-validation score. Similar symbols in the STF section—for example Q_{ab} and J_Q —must not be confused with that separate reporting layer.

9.3 Claim audit

Table 9: Admissible and inadmissible conclusions.

Statement	Status	Boundary
The registered MES ceilings obey Eqs. (11) and (12)	Supported	Explicit premises and non-negative branch
The MES transform adds independent information to C_2, C_3	Rejected	Invertibility, Eq. (16)
The scalar MES-squared family has legacy rank 98/301	Supported	Paired finite pool; absolute-median reducer
The all-two-sided ECDF companion rank is 148/301	Supported	Same rows; reducer sensitivity, not a new sky
The physics-oriented ECDF tail companion is 93/301	Supported	Same rows and cell; predeclared tail sensitivity
The paired frame-free STF family has rank 27/301 under both reducers	Withdrawn	Faulty orthonormal-real to complex-component adapter
The CMB-only frame-free ranks are 61/1000 and 55/1000	Withdrawn	Same adapter defect; noisy observation and noise-free null are not exchangeable
The PR3 component and external-holdout STF ranks measure cross-product stability	Withdrawn	Faulty projection and unmatched product-specific nulls
The fixed 13-template foreground-bank rank 144/301 disfavors foreground projection	Withdrawn	Faulty projection and no demonstrated sensitivity to subdominant or mixed residuals
The carrier-injection power curves characterize the registered alternatives	Withdrawn	Energy metric, rotations, and irrep extraction consume the wrong carrier convention
The intervention calibrates power of the original 301/1000 observation tests	Rejected	Different reference size, role graph, and no observed row
Algebraic carrier injections measure sensitivity to physical shear, vorticity, or a Bianchi solution	Rejected	No authorized physical response or physical template
All foreground explanations are ruled out	Rejected	Incomplete bank; no mixtures, nonlinear leakage, or causal source identification
The lower CMB-only fractions independently confirm an anomaly	Rejected	Same observation and family; different null pool
The observed tensors measure physical shear or vorticity	Rejected	Observer-space harmonic tensors only
Planck alone identifies local boost versus global tilt	Rejected	No independent depth response; rank deficient
A Bianchi family or physical source is identified	Rejected	No admitted physical response or native morphology atlas
The finite ranks are unconditional p -values or posteriors	Rejected	Null exchangeability and fidelity remain conditional

Statement	Status	Boundary
The registered eight orbit coordinates are globally complete	Rejected	A generic joint $\ell = 2, 3$ orbit has dimension nine, Eq. (6)
Every noncyclic Krylov stratum requires typed absence	Rejected	Once $\hat{Q}$ and $\hat{O}$ exist, the registered determinant is defined and equals zero; record the stratum separately

10 Limitations and next evidence requirements

Central limitation. The strongest present data result is the conditional scalar MES-plus-morphology finite-pool diagnostic. The observer-space lane is withdrawn. Increasing exactness of row identity or numerical replay cannot substitute for a correct representation adapter, a faithful null, or a missing physical response.

1. **Representation validity.** The orthonormal real carrier must be converted by Eq. (3) before the STF formulae are evaluated. Rotation-covariance tests and the exact radial identities in Eqs. (4)–(5) are mandatory acceptance tests for any rerun.
2. **Catalogue sufficiency.** Eight smooth invariants cannot separate the generic nine-dimensional joint orbit. Any replacement must handle continuous rank and discrete sign, permutation, and parity identifications rather than merely appending a convenient coordinate.
3. **Null fidelity.** The retained scalar paired pool is CMB plus admitted noise and remains conditional on FFP10 fidelity. The old CMB-only, product-replacement, and holdout pools are not admitted calibrations.
4. **Foreground and power scope.** No foreground falsification or method-power result is currently admitted. A replacement requires a corrected operator and a design with demonstrated sensitivity to the alternatives actually named; physical claims require physical templates.
5. **Finite resolution and dependence.** Ranks lie on finite grids, and coordinates are dependent. Participation ratios are diagnostics, not multiplicity-free physical dimensions.
6. **Premise dependence.** The principal MES numbers use the geodesic, linear almost-EGS, registered-frame, residual- $\epsilon_1 = 0$ branch. Other branches require their own recovered coefficients and assumptions.
7. **No physical response.** A future claim about shear, vorticity, local/global origin, or geometry requires a content-defined forward model, compatible conventions, rank analysis, and independent observational channels.
8. **No family identification.** A scalar rank, a sky axis, and an observer-space STF orbit are each insufficient. Family identification additionally requires a validated physical morphology atlas and null, mask, covariance, and equivalence-class gates.

The shortest scientifically meaningful next step is a code-level convention repair followed by a complete rerun of every affected row under unchanged scientific choices. Only after those semantic gates pass do matched nulls, foreground sensitivity, a separating orbit catalogue, and eventually a physical response become interpretable extensions. None is silently credited as completed here.

11 Conclusion

The current MES-bound framework is an explicit, typed chain rather than a scalar-to-physics shortcut. Low-order temperature powers yield one-way, premise-conditioned squared ceilings. Those ceilings cannot create direction or independent information. Directional morphology would have to be carried by correctly decoded harmonic blocks; no such observer-space result is admitted in this revision. A physical interpretation remains a still more separate inverse problem whose response and rank are unavailable.

On the paired 300-row primary, the scalar MES-squared family has legacy rank 98/301 and all-two-sided ECDF rank 148/301; the physics-oriented ECDF tail companion is 93/301, and the

two MES anchors are individually 74/301. The external review changed the scientific result set materially. The exact identities $q_2 = (75/8\pi)C_2$ and $o_2 = (245/8\pi)C_3$ contradict the previously reported radial values and expose a carrier/STF convention mismatch. All downstream Q/O ranks, cross-product comparisons, foreground scores, and injection-power statements are withdrawn pending full corrected rerun. The eight-coordinate family is independently rejected as a generic separating catalogue because the quotient has dimension nine.

The admitted results therefore establish only premise-conditioned MES ceilings and scalar finite-pool coordinate/reducer sensitivity. They do not establish a detection, foreground validation, an observer-space tensor anomaly, method power, a physical kinematic state, a local/global origin, or a Bianchi family.

A Explicit harmonic–STF normalization

For the quadrupole define

$$A = \sqrt{\frac{5}{16\pi}}a_{20}, \quad C = \sqrt{\frac{15}{32\pi}}, \quad D = \sqrt{\frac{15}{8\pi}}.$$

The symbols $\Re a_{\ell m}$ and $\Im a_{\ell m}$ below are the unscaled complex-harmonic components. If the input is the orthonormal real carrier, Eq. (3) must be applied first. The registered tensor is

$$Q = \begin{pmatrix} -A + 2C\Re a_{22} & -2C\Im a_{22} & -D\Re a_{21} \\ -2C\Im a_{22} & -A - 2C\Re a_{22} & D\Im a_{21} \\ -D\Re a_{21} & D\Im a_{21} & 2A \end{pmatrix}.$$

It is real, symmetric, and trace-free.

For the octupole let

$$E = \sqrt{\frac{7}{16\pi}}, \quad F = \sqrt{\frac{21}{64\pi}}, \quad G = \sqrt{\frac{105}{32\pi}}, \quad H = \sqrt{\frac{35}{64\pi}}.$$

Seven independent symmetric components are

$$\begin{aligned} O_{xxx} &= 2F\Re a_{31} - 2H\Re a_{33}, & O_{xyy} &= -\frac{2}{3}F\Im a_{31} + 2H\Im a_{33}, \\ O_{xxz} &= -Ea_{30} + \frac{2}{3}G\Re a_{32}, & O_{xyy} &= \frac{2}{3}F\Re a_{31} + 2H\Re a_{33}, \\ O_{xyz} &= -\frac{2}{3}G\Im a_{32}, & O_{yyy} &= -2F\Im a_{31} - 2H\Im a_{33}, \\ O_{yyz} &= -Ea_{30} - \frac{2}{3}G\Re a_{32}. \end{aligned}$$

Full symmetry supplies all permutations and trace-freedom fixes

$$O_{xzz} = -O_{xxx} - O_{xyy}, \quad O_{yzz} = -O_{xyy} - O_{yyy}, \quad O_{zzz} = -O_{xxz} - O_{yyz}.$$

B Compact glossary

Term	Meaning
MES ceiling	One-way premise-conditioned upper coordinate for a squared Hubble-normalized invariant
Observer-space	Constructed from measured or simulated sky harmonics; not a physical spacetime source state

Term	Meaning
STF tensor	Symmetric trace-free Cartesian representative of a fixed angular multipole
Local rank	Observation-inclusive tail rank for one row and one coordinate
Family rank	Rank of each row’s most extreme coordinate after the same scan is applied to every row
Conditional null	The specific admitted simulation pool; not an unconditional model of all uncertainty
Frame-free family	The withdrawn eight-coordinate joint $\ell = 2, 3$ catalogue; proper-rotation invariant but demonstrably not generically separating
Galactic companion	The withdrawn frame-free family plus two powers relative to the Galactic pole; no current directional inference
Method power	Rejection frequency for a correctly represented specified alternative and scoring functional; no such result is presently admitted
Typed absence	Explicitly unavailable normalized object, distinct from a defined zero on a noncyclic determinant stratum
Identified set	The range of physical states compatible with a rank-deficient response, rather than an unsupported point estimate

C Evidence and identity statement

The retained scalar result uses one observation and 300 exact ordered paired CMB-plus-noise rows. Its reported ranks, local coordinate table, family membership, tails, and reducers are unchanged by this report revision. No map was reopened and no numerical expectation or tolerance was changed. The new normalization audit is an algebraic consistency check between those same-pass C_2, C_3 values and the printed STF normalization.

The withdrawn analyses still have exact execution records, row identities, artifact hashes, and successful map-free replay. Those facts retain their proper provenance role: they show which bytes and operations produced the old objects. The external review established that the objects instantiate the wrong carrier interpretation. Their prior PASS receipts and P0/P1 counts are therefore historical process evidence, not current scientific admission. Result validity is withdrawn while source/execution provenance remains intact. No archive or hash is deleted or relabelled as a corrected result.

The decisive evidence is semantic rather than cryptographic. The carrier contract, the STF formula domain, Eqs. (4)–(5), and a true stored-basis rotation disagree with the old projection. A self-inverse map under one internally inconsistent convention cannot override those independent invariants. This is precisely the typed-identity distinction: byte identity passed, while scientific equivalence to the intended operator failed.

Identity is typed. Immutable scientific inputs and deterministic numerical tables require their declared exact content identities. Numerical arrays are accepted by their predeclared structural, residual, and numerical criteria. PDF serialization is packaging: a changed PDF byte stream is not by itself a changed rank or scientific result. Result validity, provenance validity, and packaging identity are therefore assessed separately. No tolerance, tail, feature family, or expected result was changed in preparing this report.

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